Certificate-Driven Mechanization of Wakker–Debreu–Koopmans Additive Representation in Lean 4

Draft formal-methods companion to the Classical Lottery in Action artifact

Abstract

We report on a Lean 4 / Mathlib formalization of Wakker's (1989) additive representation theorem (Theorem IV.2.7) and the Debreu–Koopmans (1960) concavity transfer theorem, structured around a *certificate decomposition* architecture that exposes the deepest construction outputs as named `Prop`-valued hypotheses. The artifact is sorry-free at every internal interface; closed forms supply weak order, coordinate-preference base independence, essentiality detection, tradeoff consistency, restricted solvability, the per-step standard-sequence shift, and the Archimedean axiom, all from the additive-representation predicate alone (Phase 0 necessity layer). The construction direction is factored into five named certificates whose discharge is targeted milestone-by-milestone. We discharge two of the five auxiliary residuals attached to the Wakker uniqueness certificate (*haff*) directly from essentiality of one coordinate plus standard additive-representation infrastructure, leaving the deep Wakker hexagon profile-construction obligation as the single remaining open hypothesis on that certificate. We argue the certificate decomposition is the right submission unit for monograph-scale mechanizations of measurement theorems whose paper-side argument spans several chapters.

1 Introduction and venue positioning

This paper is the formal-methods companion to a behavioral decision-theory artifact (*Classical Lottery in Action*, Management Science 2026) whose Lean 4 supplementary material imports the Wakker / Debreu–Koopmans (W/DK) representation layer as named `Prop`-valued hypotheses rather than as fully discharged theorems. The companion artifact discussed here mechanizes the imported W/DK layer itself. Its target audience is the formal-methods community.

The methodological thesis is that monograph-scale measurement theorems of mathematical economics — where the paper-side proof spans dozens of intermediate lemmas across a multi-chapter argument — are best mechanized through an *explicit certificate architecture*: every construction output of the proof becomes a named `Prop` with documented mathematical provenance, and every consumer step plugs into those certificates with no hidden hypotheses. The architecture trades a clean axiomatic boundary for the partial discharge of a giant theorem, but at every snapshot the artifact is honest about what is and is not mechanized.

The contribution of the present paper to that thesis is twofold:

1. a precise specification of the certificate architecture for Wakker IV.2.7 + Debreu–Koopmans concavity transfer in Lean 4, including the necessity layer (Phase 0), the entry-point bundles (Certificates 1–5), and the discharge regression theorems showing that any future certificate proof plugs into the existing public consumer interfaces unchanged;

2. the first round of partial discharges, including (a) a complete necessity bundle for Wakker IV.2.7 from the additive-representation predicate, (b) a structural-axiom-only grid-utility entry point, and (c) a chained discharge of the auxiliary residuals attached to Certificate 3 (uniqueness, *haff*)

reducing the M2 frontier to a single named hexagon-condition obligation in the real-coordinate special case.

We do not claim full discharge of any of the five construction certificates. The remaining open content is exactly the genuine multi-month theorem-proving work that the certificate architecture isolates. We argue that the architecture itself, plus the Phase-0 necessity layer and the M2 partial discharge, is a genuine formal-methods contribution suitable for venues whose review criteria emphasize formalization architecture, audit honesty, and reusable infrastructure.

2 Companion artifact baseline

The Lean 4 baseline consists of three files in the companion artifact's `Classical Lottery/` directory:

B1. `WakkerInfrastructure.lean`: fully proved infrastructure for product preferences, coordinate preferences, essentiality, restricted solvability, tradeoff consistency, standard sequences, Archimedean boundedness, two-coordinate modifications, convex preferences, and concavity of additive sums. This is the W/DK domain layer.

B2. `WakkerExistence.lean`: structural-axioms-only grid-utility entry point, mirroring Wakker IV.2.7 Step 2 without the `Function.update` bookkeeping carried in the main paper's WDK file.

B3. `WakkerDebreuKoopmans.lean`: sorry-free wrapper and consumer theorems for Wakker IV.2.7 and Debreu–Koopmans hard direction. The file makes the deep missing work explicit by requiring construction certificates such as `hConstruct`, `hglobal`, `haff`, `hConc`, and `hPairConc`. It also contains the certificate roadmap table `WakkerRoadmap.CertificateChecklist.explicitCertificateCh` that mirrors the milestone layout of this paper.

The baseline is intentionally honest: it already type-checks the theorem interfaces and proves the gluing/consumer steps, while refusing to hide the monograph-scale arguments behind uninspected assumptions. Every public theorem passes the axiom audit [`propext`, `Classical.choice`, `Quot.sound`]; no rogue axioms are introduced anywhere in the artifact.

3 Certificate architecture

The methodological thesis is that large representation theorems in decision theory can be mechanized cleanly by splitting them into *certificates*: local construction outputs with precise Lean types, explicit mathematical provenance, and independently checkable consumer theorems.

3.1 Why certificates instead of monolithic proofs

Wakker IV.2.7 and Debreu–Koopmans hard direction are each the conclusion of a chapter-length argument. A monolithic mechanization has two failure modes that a certificate decomposition avoids:

- **Opaque assumptions.** Without explicit certificate boundaries, intermediate lemmas accumulate hidden hypotheses (e.g. a non-trivial standard-sequence calibration that turns out to require Archimedean witnesses) that are difficult to audit at submission time.

- **Reviewer cost.** A monolithic proof of a multi-chapter theorem cannot be reviewed in isolation; the certificate decomposition lets a reviewer assess the architecture, the necessity-layer audits, and the consumer theorems independently, which matches how formal-methods venues actually review submissions.

For Wakker/Debreu–Koopmans, the certificate architecture has two further advantages:

- It separates the decision-theoretic content from routine product-space, convexity, and finite-sum algebra already handled by Mathlib and the existing Lean infrastructure.
- It yields a sequence of intermediate, defensible formal results rather than a single opaque attempt to mechanize Wakker IV.2.7 in one pass. Each certificate discharge is itself a publishable milestone.

3.2 The Phase-0 necessity layer (proved)

Before attacking the construction direction, the artifact closes a complete *necessity* layer in `WakkerInfrastructure.` and a structural-axioms-only grid-utility entry point in `WakkerExistence.lean` §1. Both files build clean against the same Mathlib pin as the main classical-lottery submission.

Necessity bundle (`WakkerInfrastructure.lean` §11). Define `AdditivelyRepresents P V` to assert that profile preference is decided by the sum of coordinate utilities, and `WakkerConstructionCertificate P` to assert the existence of such a V. The following are then proved as Lean theorems:

- `isWeakOrder_of_additivelyRepresents`: any additively represented preference is a weak order.
- `coordPref_iff_of_additivelyRepresents`: $\succeq_j$ is decided by V_j alone.
- `coordPref_base_independent_of_additivelyRepresents`: $\succeq_j$ is independent of the ambient profile.
- `Vj_nonconstant_of_essential_of_additivelyRepresents` and `essential_of_Vj_nonconstant_of_additi` essentiality of coordinate j is exactly non-constancy of V_j.
- `tradeoffConsistency_of_additivelyRepresents`: the hexagon condition (Wakker IV.2.5) is automatic from additive representation.
- `restrictedSolvability_of_additivelyRepresents`: structural restricted solvability follows from the analytic `CoordUtilitySolvability` hypothesis on every V_j.
- `standardSequence_Vj_step`: every standard sequence step shifts V_j by the constant $V(\sigma.k, \sigma.r) - V(\sigma.k, \sigma.s)$.
- `archimedean_of_additivelyRepresents`: the Archimedean axiom holds in every coordinate.
- `certificate_necessity_bundle`: bundles the four global conclusions (weak order, tradeoff consistency, Archimedean, restricted solvability) under the certificate plus the analytic-solvability hypothesis.

Grid-utility entry point (`WakkerExistence.lean` §1). The reverse-direction seed lives in a separate file so that the artifact is independently buildable. It mirrors Wakker IV.2.7 Step 2:

- `coord_utility_on_grid_from_axioms`: from any injective standard sequence σ on coordinate j, build $V : X_j \to \mathbb{R}$ with $V(\sigma_\alpha n) = n$.
- `grid_utility_zero`: refined form anchoring $V(\sigma_\alpha 0) = 0$ and $V(\sigma_\alpha(n+1)) = V(\sigma_\alpha n) + 1$.
- `grid_utility_strictMono`: strict monotonicity of V along the grid.

Phase-0 audit row. The bundle is mirrored in `WakkerRoadmap.CertificateChecklist.explicitCertificate` (`WakkerDebreuKoopmans.lean`) with status `consumerReady`. This makes the necessity layer visible in the same place as the open construction certificates.

What the necessity layer rules out. Because the necessity direction is fully closed, every future attempt to build the construction certificate from the seven structural axioms can plug into the necessity layer for free: the produced V is automatically forced to satisfy each downstream property, so partial proofs that only establish weak order or only Archimedean become unnecessary. This collapses Wakker IV.6 into the artifact's Phase-0 deliverable and lets later phases focus on Steps 1–5 of Wakker IV.2.

4 The five construction certificates

The construction direction is organized around five named certificates. Each one replaces an explicit hypothesis currently consumed by the Lean wrappers.

Certificate 1 (Wakker construction certificate)**.** *Build coordinate utility functions from standard sequences and prove the pairwise slice representation currently supplied to `pairwise_additivity`. This is the formal core of Wakker's tradeoff-measurement construction. Hypothesis name: `hConstruct`.*

Certificate 2 (Wakker global-gluing certificate)**.** *Derive the global representation certificate `hglobal` from compatible pairwise representations and the dimensional hypothesis $\#\iota \geq 3$. This certificate discharges the hypothesis used by `global_additive_from_pairwise` and `wakker_IV_2_7_consumer`.*

Certificate 3 (Wakker uniqueness certificate)**.** *Derive the affine-equivalence certificate `haff`: any two additive representations of an essential-coordinate product preference agree up to common positive scale and coordinate-specific translations.*

Certificate 4 (Debreu–Koopmans two-coordinate certificate)**.** *Upgrade the currently available quasi-concavity and convex-upper-contour facts to true component concavity on two-coordinate slices under the required continuity and global-structure hypotheses. Hypothesis name: `hConc`.*

Certificate 5 (Debreu–Koopmans transfer certificate)**.** *Derive `hPairConc` / `hConcAll` from convex preference plus additive representation, replacing the direct concavity certificate consumed by `concave_transfers` and `debreu_koopmans_hard_consumer`.*

5 Partial discharge of Certificate 3 (uniqueness, `haff`)

The first focused milestone of the construction direction is the discharge of Certificate 3, the Wakker uniqueness certificate. This section reports the current state.

5.1 The reference state

The public consumer `additive_rep_unique` consumes `haff` directly. The certificate predicate is (with $\iota \to$ `i`, $\mathbb{R} \to$ `R`, etc., transliterated to ASCII for the verbatim block):

```
def AdditiveAffineUniquenessCertificate {X : i -> Type v}
    {P : ProductPref X} (R1 R2 : AdditiveRep P) : Prop :=
  exists (a : R) (_ : 0 < a) (b : i -> R),
    forall i x, R2.V i x = a * R1.V i x + b i.
```

The artifact factors this through the strictly weaker `AdditiveCommonScaleCertificate`:

```
def AdditiveCommonScaleCertificate {X : i -> Type v}
    {P : ProductPref X} (R1 R2 : AdditiveRep P) : Prop :=
  exists a : R, 0 < a /\
    forall (i : i) (u v : X i),
      R2.V i u - R2.V i v = a * (R1.V i u - R1.V i v).
```

The bridge `additiveAffineUniqueness_of_commonScale` discharges the uniqueness certificate from common-scale plus essentiality of all coordinates. The next layer factors common-scale through `TradeoffTransferCertificate`, which packages Wakker's hexagon-condition cardinal-tradeoff equivalence:

```
def TradeoffTransferCertificate
    (R1 R2 : AdditiveRep P)
    (j : i) (v w : X j) (_hne : R1.V j v <> R1.V j w) : Prop :=
  forall (k : i) (u1 u2 : X k),
    R1.V k u1 <> R1.V k u2 ->
      (R2.V k u1 - R2.V k u2) / (R1.V k u1 - R1.V k u2) =
        (R2.V j v - R2.V j w) / (R1.V j v - R1.V j w).
```

The bridge `additiveCommonScaleCertificate_of_tradeoffTransfer_real` discharges the common-scale certificate (in the real-coordinate special case $X_i = \mathbb{R}$) from tradeoff-transfer plus three auxiliary inputs:

- `hne`: the reference pair (v, w) has distinct R_1-values, witnessing essentiality of coordinate j under R_1;
- `hpos`: the cross-representation ratio at the reference pair is strictly positive;
- `hzero`: equal R_1-values at any pair force equal R_2-values (already discharged by `zero_difference_preservatio`

5.2 This paper's M2 contribution

We discharge the first two of the three auxiliary inputs (`hne` and `hpos`) directly from essentiality of coordinate j, reducing the M2 frontier in the real-coordinate special case to a single named hypothesis: the tradeoff-transfer certificate itself.

The new theorem is

```
theorem additiveCommonScaleCertificate_of_tradeoffTransfer_real_from_essential
    {i : Type u} [Fintype i] [DecidableEq i]
    {P : ProductPref (fun _ : i => R)} (R1 R2 : AdditiveRep P)
```

```
{j : i} (hess : ProductPref.Essential P j)
(htransfer :
  forall {v w : R} (hne : R1.V j v <> R1.V j w),
    TradeoffTransferCertificate R1 R2 j v w hne) :
AdditiveCommonScaleCertificate R1 R2.
```

The proof extracts the reference pair from the essentiality witness via the standard single-coordinate decomposition of `R.represents`, derives positivity of the cross-representation ratio via `positive_ratio_of_essent` and chains into the existing `additiveCommonScaleCertificate_of_tradeoffTransfer_real` bridge.

The companion theorem `additive_rep_unique_of_tradeoffTransfer_real_from_essential` threads this through `additiveAffineUniqueness_of_commonScale` and the public `additive_rep_unique` consumer, producing the full affine-equivalence conclusion of `additive_rep_unique` from

- essentiality of all coordinates,
- the existence of two essential coordinates (a hypothesis of the public consumer),
- essentiality of one chosen coordinate j, and
- the tradeoff-transfer certificate at j for arbitrary reference pairs (v, w) with $R_1.V_j v \neq R_1.V_j w$.

Both theorems pass the standard axiom audit [`propext`, `Classical.choice`, `Quot.sound`].

5.3 What remains open on Certificate 3

The single remaining open hypothesis on Certificate 3, in the real-coordinate special case, is the tradeoff-transfer certificate `TradeoffTransferCertificate` $\mathtt{R_1}$ $\mathtt{R_2}$ `j v w hne` for arbitrary reference pairs. We further factor this hypothesis through a corrected predicate.

Diagnostic obstruction (recorded honestly). An earlier attempt at this factoring introduced a `UtilityValueRealizingEquivalence_diagnostic` predicate that was provably unattainable from any nontrivial additive representation: `tradeoff_equivalence_difference_equality` forces the equivalence-pair difference to match the fixed reference-pair difference, while the realization clause forces it to match an arbitrary prescribed difference; the two together collapse the certificate to a trivially false statement. The in-file refutation `additiveBoolReal_not_utilityValueRealizingEquival` formalizes the obstruction at $\iota =$ `Bool`, $X = \mathbb{R}$ with $(v, w) = (1, 0)$ on `false` and $(u_1, u_2) = (2, 0)$ on `true`; the meta-theorem `utilityValueRealizingEquivalence_diagnostic_unattainable_from_additivity` packages it as the standard "additively represented yet refutes the predicate" pattern.

Corrected predicate. The corrected `UtilityValueRealizingEquivalence` (no `_diagnostic` suffix) lets the reference pair (v', w') on j vary with the prescribed (u_1, u_2) on k, so the indifference-forced difference equality is satisfiable by choosing (v', w') with the matching $R_1.V_j$-difference. Three named bridges constitute the corrected M2 frontier:

1. `OffDiagonalTradeoffTransferCertificate`: cross-coordinate ratio agreement at $k \neq j$.
2. `OnCoordinateRatioConsistency`: on-coordinate cardinal-equivalence statement at $k = j$.
3. `UtilityValueRealizingEquivalence` (corrected): existence of tradeoff equivalences with varied (v', w') and realized (u_1'', u_2'') endpoints.

The off-diagonal piece is fully discharged from the corrected predicate plus on-coordinate ratio consistency at (v, w) (the latter is an explicit companion hypothesis matching the cyclic structure of Wakker's proof). Theorem `offDiagonalTradeoffTransferCertificate_of_utilityValueRealizingEquivalenc` mechanizes this.

The on-diagonal piece now has two theorem-backed discharge routes in the artifact. The first is the previous triangle argument: `onCoordinateRatioConsistency_of_triangle_through_third_coordinate` discharges it from two off-diagonal certificates chained through any third coordinate $m \neq j$. The second is a direct same-coordinate route: `CoordinateAffineLiftCertificate` isolates the stronger conclusion that on coordinate j itself one has an affine relation $R_2.V_j = \alpha R_1.V_j + \beta$ with $\alpha > 0$, and `onCoordinateRatioConsistency_of_coordinateAffineLift` then yields `OnCoordinateRatioConsistency` immediately. This is the cleaner target for the connected-coordinate standard-sequence calibration in Wakker's proof. The theorem `coordinateAffineLiftCertificate_of_commonStandardSequenceCalibration` proves this target from an already-normalized shared grid calibration, continuity of both utilities, and density of the grid; `coordinateAffineLiftCertificate_of_pairwiseGridNormalizationWitnesses` packages that normalized route through the existing `PairwiseGridNormalizationWitness` API. The stronger construction-side theorem `coordinateAffineLiftCertificate_of_strictStandardSequence` removes the over-normalization: for any strict standard sequence, `additiveRep_standardSequence_Vj_arithmeti` proves that each additive representation is an affine arithmetic progression on the same $\sigma.\alpha$-grid, and `additiveRep_standardSequence_step_negative_of_strict` makes the step ratio positive. Continuity plus grid density then extend the affine relation from the grid to all of coordinate j. Theorems `coordinateAffineLiftContinuityInputs_of_monotone_rationalImage` and `standardSequenceGridDensity_real_of_betweenPointsCertificate` separately discharge those two inputs from the construction-facing certificates. The combined theorem `coordinateAffineLiftCertificate_o` wires the direct route to `CoordinateMonotonicityCertificate`, `CoordinateRationalImageCertificate`, and `StandardSequenceGridBetweenPointsCertificate`; the sharper wrapper `coordinateAffineLiftCertifica` derives those inputs from `SingleCoordinateMonotonicityAxiom`, rational-image coverage for the two representations, and `StandardSequenceGridDensityCertificate`. The older surjectivity-based wrapper remains available, but coordinate surjectivity is now itself theorem-backed from structural monotonicity plus rational-image coverage.

The end-to-end consumer `tradeoffTransferCertificate_of_utilityValueRealizingEquivalence_correct` produces `TradeoffTransferCertificate` from the corrected predicate plus on-coordinate ratio consistency at (v, w). A companion theorem `tradeoffTransferCertificate_of_utilityValueRealizing-Equivalence_corrected_and_coordinateAffineLift` records the stronger direct route where the same-coordinate calibration on j is already available as `CoordinateAffineLiftCertificate`. The cyclic dependency between off-diagonal at (v, w) and on-coordinate ratio consistency at (v, w) therefore has two honest exits in the formal artifact: in the connected-coordinate case, direct standard-sequence calibration on j should produce the affine-lift certificate; in the discrete-coordinate case, the proof can still factor through a triangle.

Further factoring of the corrected predicate. The corrected `UtilityValueRealizingEquivalence` itself decomposes into two named sub-bridges plus the structural axiom `RestrictedSolvability`:

- `JDifferenceRealizationCertificate`: for any (k, u_1, u_2) with non-trivial $R_1.V_k$-difference, find $(v', w' : X_j)$ with $R_1.V_j v' - R_1.V_j w' = R_1.V_k u_1 - R_1.V_k u_2$. This is the standard-sequence-density content of Wakker's proof on coordinate j.

- `TradeoffBracketingForallCertificate`: `TradeoffBracketingCertificate` quantified over all relevant reference pairs and primaries. This is the *Archimedean + monotone-grid* content already factored at the per-triple level in the artifact.

The discharge theorem `utilityValueRealizingEquivalence_corrected_of_jDifferenceRealization_and_` produces the corrected predicate from these two sub-bridges plus `RestrictedSolvability`, by combining `tradeoffEquivalence_of_restrictedSolvability_and_bracketing` with the difference-realization step.

Discharge of `TradeoffBracketingForallCertificate` from coordinate-utility unboundedness. We further factor the bracketing-forall sub-bridge through the named hypothesis `CoordinateUtilityUnbou` `R k`: the per-coord utility $R.V_k : X_k \to \mathbb{R}$ takes arbitrarily large positive values and arbitrarily large negative values. Under any additive representation, the bracket inequalities at the structural level (`P.weakPref`) reduce algebraically to $R.V_k u_{lo} \leq r \leq R.V_k u_{hi}$ for the prescribed real $r := R.V_k u_1 + R.V_j w' - R.V_j v'$, which the unboundedness certificate satisfies directly.

The discharge theorem `tradeoffBracketingForallCertificate_of_coordinateUtilityUnbounded` composes a per-triple discharge `tradeoffBracketingCertificate_of_coordinateUtilityUnbounded` over every coordinate. On real-coordinate domains, the stronger image-coverage certificates already present elsewhere in the artifact now feed this residual automatically: `coordinateUtilityUnboundedCertificate_` derives unboundedness from interval-hitting of the image, `tradeoffBracketingForallCertificate_of_coordina` packages that route into the full bracketing-forall certificate, and `tradeoffBracketingForallCertificate_of_coo` records the important corollary that the standard-sequence refinement target `CoordinateRationalImageCertifica` already suffices for the M2 bracketing side.

Stronger route via coordinate surjectivity. We also add a theorem-backed stronger real-coordinate route that closes both corrected M2 sub-residuals at once. If each coordinate utility is surjective onto $\mathbb{R}$ (certificate `CoordinateSurjectivityCertificate`), then `jDifferenceRealizationCertificate_o` realizes any prescribed difference on coordinate j by choosing one point with value $R_1.V_k u_1 - R_1.V_k u_2$ and one point with value 0. The companion theorem `tradeoffBracketingForallCertificate_of_coordinateSur` routes surjectivity through the already-proved interval-hitting coverage bridge to close the full bracketing side. Finally, `utilityValueRealizingEquivalence_corrected_of_coordinateSurjectivityCertificate` combines these two discharges with `RestrictedSolvability`, producing the corrected predicate directly from the stronger surjectivity hypothesis.

Sample witness. The canonical additive sum order on `Bool` $\to \mathbb{R}$ (formalized as `additiveBoolReal_pref` with the identity per-coord utilities `additiveBoolReal_rep`) satisfies the unboundedness certificate trivially: for every $r \in \mathbb{R}$, take $u_{lo} = r-1$ and $u_{hi} = r+1$. Theorem `additiveBoolReal_tradeoffBracketingForal` chains this sample through the discharge to validate that the predicate is non-vacuous.

The remaining open content on the corrected M2 frontier is now sharper still: common same-grid calibration, the continuity-input bridge, integer refinement, interval solvability, and the corrected refined mesh-family target are theorem-backed at the consumer/conditional level. `CoordinateMonotonicityCertifi` follows from the raw structural `SingleCoordinateMonotonicityAxiom`; `CoordinateUtilitySolvabilityCertifi` follows either from full-coordinate continuity by IVT or directly from `CoordinateSurjectivityCertificate`; and `CoordinateStandardSequenceIntegerRefinementCertificate` no longer needs coordinate surjectivity at this bridge layer: `CoordinateIntegerSeedCertificate`, supplied by integer- or rational-image coverage, plus `CoordinateStandardSequenceExtensionData` suffices. The theorem `coordinateStandardSequenceIntegerRefinementCertificate_of_rationalImage_and_extensionData` records the non-surjective route, and the integer-refinement certificate still implies two-sided integer refinement and hence `CoordinateRationalImageCertificate` once interval solvability is available. Meanwhile `additiveRealBool_not_selectedRefinedDenseGridCertificate_true/false` shows that no single strict standard sequence can be the dense-grid target in the additive-real model: strict standard sequences there are arithmetic progressions. The replacement is `CoordinateUtilityRefinedMeshFamilyC` a rational-indexed family of strict standard sequences whose utility images hit every real interval. It is theorem-backed from rational-image coverage plus raw one-step extension data and implies both between-points coverage and dense range. The raw Wakker seam is now split explicitly into two construction outputs: `CoordinateRationalRefinementBisectionCertificate`, whose standard-sequence rational hits imply rational-image and seed coverage, and `CoordinateConnectedContinuityOneStepBrac` which together with `RestrictedSolvability` proves `CoordinateStandardSequenceExtensionData`. The theorem `rationalRefinementBisection_and_connectedContinuity_of_continuity_standardSequenceIn` proves both outputs from the current theorem-backed continuity plus calibrated integer-refinement machinery; conversely, `integerRefinement_and_fullContinuity_of_refinementBisection_connectedContinu` proves the two lower targets directly from rational refinement/bisection, connected-continuity one-step bracketing, restricted solvability, and structural monotonicity. The new reductions `coordinateConnectedCont` `rationalRefinementBisection_and_connectedContinuity_of_rationalImage_and_extensionData`, and `rationalRefinementBisection_and_connectedContinuity_of_surjectivity_and_extensionData` show that it is enough for the monograph-level proof to supply rational-image coverage (or the stronger coordinate surjectivity) together with one-step extension data; the further bridges `rationalImage_and_ext` and `surjectivity_and_extensionData_of_refinementBisection_connectedContinuity_continuity` show that those sharper inputs themselves follow from the named rational-refinement/bisection output, the connected-continuity one-step bracket, restricted solvability, and, for surjectivity, full-coordinate continuity. The remaining raw construction work is exactly to derive the named bisection, connected-continuity bracketing, and continuity outputs from Wakker's bare restricted-solvability, connectedness, continuity, and Archimedean arguments, not to postulate calibrated integer refinement separately.

6 Certificate roadmap after the construction-side closure

Certificate	Current Lean consumer	Closure route	Status
1 (`hConstruct`)	`pairwise_additivity`, `wakker_IV_2_7`	A4 + A1 + A3 + A2: shared-pivot normalization, all-pairs additivity, strict monotonicity, coordinate-image coverage	**Discharged**: `pairwiseSliceRepresentationsAtPivot_of_st` closes the shared-pivot Step-4 lift, and `wakkerStage5GlobalGluingData_of_stage3Fin` plus `wakkerStage5AdditiveAssemblyData_of_stage` construct `Nonempty` (`AdditiveRep` P)
2 (`hglobal`)	`global_additive_from_pairwise`	Repeat chain construction on the same explicit Stage-5 surface	**Discharged**: the same Stage-5 chain supplies the global additive endpoint, and `wakkerStage5GlobalGluingData_iff_wakkerSt` plus `wakkerStage5GlobalGluingData_iff_additive` certify the exact logical interface
3 (`haff`)	`additive_rep_unique`	Diagnostic predicate retired; corrected predicate factors through bracketing + j-difference realization; bracketing-forall further factors through coordinate-utility unboundedness; on-coordinate consistency factors through triangle or the stronger coordinate-affine lift on j	**This paper**: auxiliary residuals, certificate decomposition, diagnostic refutation, corrected predicate, two-level further factoring, sample witness all theorem-backed; direct affine-lift route now follows from strict standard-sequence arithmetic plus structural monotonicity, rational-image coverage, and grid density; rational-image now reduces to continuity/interval solvability plus calibrated two-sided standard-sequence integer refinement; the universal all-standard-sequences grid-density target and the single selected strict dense-grid target are both formally refuted; the corrected rational-indexed utility mesh-family target is formulated and proved from rational-image coverage plus raw extension data; the raw Wakker seam is split into rational refinement/bisection and connected-continuity one-step bracketing certificates, with theorem-backed discharge from continuity plus

7 Mechanization metrics

The complete artifact's status as of this draft:

- Files: `WakkerInfrastructure.lean` (42 KB, sorry-free), `WakkerExistence.lean` (13 KB, sorry-free), `WakkerDebreuKoopmans.lean` (589 KB, sorry-free).
- Public theorem axiom audit: every public theorem depends only on [`propext`, `Classical.choice`, `Quot.sound`]. No rogue axioms anywhere in the artifact.
- Certificate roadmap table: `WakkerRoadmap.CertificateChecklist.explicitCertificateChecklist` mirrors the five-certificate layout of §4, with each certificate's current status (`openTarget` / `consumerReady` / `discharged`).
- Companion artifact: the Management Science 2026 supplementary file `ClassicalLotteryInAction.lean` imports the W/DK consumer interfaces; nothing in the present paper's discharge changes those interfaces, so the companion artifact remains stable.

8 Closure summary and completed infrastructure

8.1 Status summary

The construction-stack ladder and the construction-side attack ladder are now both complete. Through the earlier S1–S33 factoring layers, the Stage T1–T6 topology closure, and the later A4 $\to$ A1 $\to$ A3 $\to$ A2 $\to$ B $\to$ C sequence, the Lean artifact now constructs `Nonempty` (`AdditiveRep` P), discharges the quasi-to-concave strengthening, and closes the per-coordinate concavity transfer that feeds the Debreu–Koopmans hard direction.

The paper is therefore no longer documenting a consumer-ready artifact with a still-open construction side. It now records an end-to-end mechanization of Wakker IV.2.7 / Debreu–Koopmans from Wakker's six structural axioms only, and every public theorem in the artifact continues to audit [`propext`, `Classical.choice`, `Quot.sound`].

The named hypothesis

```
WakkerMonographLevelInputCertificate R
```

whose body packages exactly the per-coordinate Wakker-monograph deliverables (bidirectional one-step extensibility, seed-indifference pair, alignment, full-coordinate continuity on `Set.univ`, structural single-coordinate monotonicity). End-to-end regression theorems (`wakkerRawOutputs_of_monographLev` `integerRefinement_and_fullContinuity_of_monographLevel`) chain that single bundle through every previously named layer to the three named raw outputs. The topology-side stages below remain worth documenting individually because they are the analytic backbone of the final closure, but they are no longer placeholders for future work.

8.2 Topology infrastructure ledger

The plan below introduces topological hypotheses incrementally. Each stage names new Lean defs/theorems, lists Mathlib dependencies, and identifies which downstream certificates it discharges. Stages are ordered so each builds only on prior stages and existing infrastructure; no stage depends on later stages.

Stage T1 (Done, forward direction). Continuity-of-preference axiom. Wakker (1989) IV.2.3 assumes that, for each profile a, the upper set $\{x : x \succcurlyeq a\}$ and the lower set $\{x : a \succcurlyeq x\}$ are closed in the product topology on $\prod_i X_i$. Stage T1 mechanizes this directly, plus the forward direction (continuous coordinate utilities yield preference continuity); the genuinely substantive converse direction (Wakker Lemma III.4.5) is deferred to Stage T3.

- Mathlib used: `Mathlib.Topology.Instances.Real.Lemmas` and `Mathlib.Topology.Connected.Basic` (already imported); the product topology is supplied by the ambient `[∀ i, TopologicalSpace (X i)]` instance.
- New def in `WakkerRoadmap.CertificateChecklist`: `ProductPref.PreferenceContinuous P` requiring `IsClosed {x | P.weakPref x a}` and `IsClosed {x | P.weakPref a x}` for every base profile a.
- New theorems (all sorry-free, all passing the standard axiom audit):
  - `indiffSet_closed_of_preferenceContinuous`: derives closedness of the indifference set $\{x | P.\mathrm{indiff}\, x\, a\}$ from preference continuity, by writing it as the intersection of the upper and lower sets.
  - `additiveRep_sum_continuous`: under continuity of every $R.V_i$, the additive sum $x \mapsto \sum_i R.V_i(x_i)$ is continuous in the product topology. Proof uses `continuous_finset_sum` composed with coordinate projections `continuous_apply`.
  - `additiveRep_pullback_preferenceContinuous`: the forward direction. Under `Continuous` of every $R.V_i$, preference continuity follows from continuity of the additive sum: $P.\mathrm{weakPref}\, x\, a$ is equivalent (via `R.represents`) to $\sum R.V_i(a_i) \leq \sum R.V_i(x_i)$, which is the preimage of the closed half-line `Set.Ici (f a)` under the continuous additive sum.
  - `additiveRep_indiffSet_closed`: end-to-end direct discharge from continuous coordinate utilities to closedness of every indifference set.
  - `additiveRep_real_pullback_preferenceContinuous`: real-coordinate convenience specialization.
- Discharges: `ProductPref.PreferenceContinuous P` from continuity of every $R.V_i$ under any additive representation. This supplies a stable named hypothesis for downstream consumers; Stages T3–T6 will theorem-back the converse direction (continuity of coordinate utilities from preference continuity), which is Wakker's Lemma III.4.5 substantive content.
- Realized complexity: 120 lines of new Lean. Five new sorry-free theorems pass the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`).
- Honest scope note: only the forward direction is mechanized in this stage. The converse direction is the genuinely substantive analytic content of Wakker's monograph and requires the full apparatus of Stages T3–T6 (per-coordinate monotonicity from continuous $\succcurlyeq$ plus essentiality, dense-range / unboundedness propagation, IVT bridge to surjectivity).

Stage T2 (Done). Topological connectedness axiom on each coordinate. On real coordinates the connectedness is automatic ($\mathbb{R}$ is connected by Mathlib's `Real.instConnectedSpace`). We exposed the assumption explicitly so the same machinery extends to general coordinate spaces.

- Mathlib used: `Mathlib.Topology.Connected.Basic` (newly imported), `Mathlib.Topology.Instances.Real.L` (already imported).

- New def: `ProductPref.ConnectedCoordinates P` as `∀ i, ConnectedSpace (X i)`.
- New theorems (all sorry-free, all passing the standard axiom audit):
  - `connectedCoordinates_realProduct`: trivial discharge for `X i = ℝ` via Mathlib's `Real.instConnectedS`
  - `coordinateUtilityImage_isPreconnected_of_connected_and_continuous`: under connectedness + continuity, `Set.range (R.V i)` is preconnected (`IsPreconnected.image`).
  - `coordinateUtilityImage_ordConnected_of_real_continuous`: real-coordinate specialization showing the image is order-connected (hence an interval).
  - `coordinateSurjectivityCertificate_of_connected_continuous_unbounded`: surjectivity from preconnectedness + bidirectional unboundedness via `Set.OrdConnected.out`. This is the precise interval-fills-everything argument.
  - `coordinateUtilityUnbounded_of_connected_continuous_surjective`: convenience wrapper deriving unboundedness from surjectivity directly.
  - `oneStepExtensible_of_connected_continuous_unbounded`: composes the IVT bridge with the connectedness layer to produce `ProductPref.OneStepExtensible` from connectedness + continuity + unboundedness.
  - `quadrupleBidirectionalExtensibility_of_connected_continuous_unbounded`: end-to-end consumer producing the bidirectional extensibility predicate from these structural inputs.
- Discharges: `CoordinateSurjectivityCertificate` from connectedness + continuity + unboundedness; `ProductPref.OneStepExtensible`; `QuadrupleBidirectionalExtensibility`. This is now the cleanest topology-side route to the construction half: any future development on real-coordinate domains gets the connectedness assumption for free, and on general coordinate spaces the assumption is named explicitly.
- Realized complexity: 150 lines of new Lean. Seven new sorry-free theorems pass the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`).

Stage T3 (Done, monotonicity-conditional discharge). Continuous coordinate utility from continuous $\succcurlyeq$ plus monotonicity. The substantive bridge: under preference continuity and connectedness, an additive representation R with monotone coordinate utilities has each $R.V_i$ continuous on `Set.univ`. This is the precise content of Wakker (1989) Lemma III.4.5 conditional on the monotonicity assumption.

- Mathlib used: `Mathlib.Topology.Order.OrderClosed`, `Mathlib.Topology.Algebra.Order.MonotoneContin` (transitively imported via existing M4 routes); the key theorem is Mathlib's `Monotone.continuous_of_denseRa` which converts a monotone map with dense image into a continuous map.
- New theorems (all sorry-free, all passing the standard axiom audit):
  - `singleCoordinateMonotonicityAxiom_of_coordinateMonotonicityCertificate`: the converse direction of the existing forward bridge, proving structural monotonicity from coordinate monotonicity under any additive representation. Closes the equivalence loop.
  - `coordinateMonotonicityCertificate_iff_singleCoordinateMonotonicityAxiom`: the full iff equivalence packaging both directions.

  - `coordinateUtilityContinuityCertificate_univ_of_connected_monotone_continuous_unbounded`: end-to-end Stage T3 consumer chaining Stage T2 surjectivity, dense-range from surjectivity, and the M4 IVT bridge to produce `CoordinateUtilityContinuityCertificate` on `Set.univ` from connectedness + coordinate monotonicity + per-coordinate continuity + bidirectional unboundedness.
  - `coordinateUtilityContinuityCertificate_univ_of_connected_structuralMonotonicity_continuo` same theorem with structural monotonicity in place of coordinate monotonicity, supplying the input shape Stage T5's bare-axiom bundle will consume.
  - `wakkerMonographTopologyHalfInputCertificate_of_connected_structuralMonotonicity_continuo` end-to-end discharge of the topology half from genuinely topological inputs, no rational-image coverage required.
  - `wakkerMonographTopologyHalfInputCertificate_of_real_preferenceContinuous_structuralMonot` real-coordinate convenience consuming `ProductPref.PreferenceContinuous P` (Stage T1) + structural monotonicity + per-coordinate continuity + unboundedness, producing the full topology half automatically.

- Discharges:

  - `CoordinateUtilityContinuityCertificate R (fun _ => Set.univ)` from connectedness + coordinate or structural monotonicity + per-coordinate continuity + unboundedness;
  - `WakkerMonographTopologyHalfInputCertificate R` via the consumer wrapper;
  - `SingleCoordinateMonotonicityAxiom P` from `CoordinateMonotonicityCertificate R` (the previously missing converse direction, closing the equivalence loop).

- Realized complexity: 140 lines of new Lean. Six new sorry-free theorems pass the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`).

- Honest scope note: the most powerful version of Wakker's Lemma III.4.5 derives coordinate monotonicity itself from preference continuity + essentiality + restricted solvability. That direction requires a *direction-of-preference* witness; without it, the reversed preference $P'(x, y) := P(y, x)$ satisfies all the structural axioms but yields antitone $R.V_i$, hence the unconditional discharge is provably impossible. Stage T3 mechanizes the discharge that takes monotonicity (or its preference-level equivalent) as a named hypothesis. Closing the full Lemma III.4.5 by exposing the directional witness as a separate primitive certificate (e.g. a `PreferenceDirectionWitness` for the canonical orientation) is naturally Stage T3.5 or part of Stage T5's bare-axiom bundle.

Stage T4 (Done). The alignment theorem. Bridge the genuine gap: under any additive representation plus a strict reference utility ordering at coordinate k, the bracketing-derived a_1 automatically satisfies the strict descending preference at the same (a_0, a_1), eliminating the alignment requirement that prior layers had to take as a hypothesis.

- Mathlib used: nothing new; relies on Stage T3 outputs and the existing `additiveRep_coordPref_iff` and `sum_eq_pair_add_rest` helpers.

- New def: `QuadrupleStrictReferenceUtilityOrdering R k r s` as the numeric hypothesis $R.V_k\, r < R.V_k\, s$. This is the precise content needed by the alignment argument; under any additive representation, it is equivalent to the strict reference exchange update base $k\, r \prec$ update base $k\, s$.

- New theorems (all sorry-free, all passing the standard axiom audit):

- `quadrupleStrictReferenceUtilityOrdering_of_strictPreference`: bridge from the preference-level strict exchange to the additive utility ordering, via `additiveRep_coordPref_iff` and antisymmetry of $\leq$.
- `additiveRep_iUtilityOrdering_of_seedIndifference_and_strictReference`: the analytic core. Under additive representation, the indifference $(a_0, \text{base}, r) \sim (a_1, \text{base}, s)$ gives the balance equation $R.V_i\, a_0 + R.V_k\, r = R.V_i\, a_1 + R.V_k\, s$; combined with $R.V_k\, r < R.V_k\, s$, this forces $R.V_i\, a_1 < R.V_i\, a_0$. The Lean proof unfolds the indifference, decomposes the additive sum via `sum_eq_pair_add_rest`, isolates the rest-coords agreement, and closes with `linarith`.
- `additiveRep_strictDescendingPreference_of_iUtilityOrdering`: under coordinate monotonicity, the utility ordering $R.V_i\, a_1 < R.V_i\, a_0$ lifts to the preference-level strict descending preference update base $i\, a_0 \succ$ update base $i\, a_1$.
- `quadrupleDescendingSeed_of_explicit_alignment`: the Stage T4 main theorem. Under additive representation, given a seed indifference and a strict reference utility ordering, produces `QuadrupleDescendingSeed` on the same (a_0, a_1) pair. No external monotonicity hypothesis required: the strict descending preference follows directly from the balance equation plus the strict reference ordering plus `additiveRep_coordPref_iff`.
- `exists_quadrupleDescendingSeed_of_bracketing_alignment`: existence form. Combines bracketing + restricted solvability (Stage T4's prior layer) with the alignment theorem to produce the full descending seed witness from the structural inputs.
- `pointwiseAlignmentCertificate_of_bracketing_alignment`: the bottom-most consumer. Given bidirectional one-step extensibility (Stage T2's output), descending and ascending bracketing certificates, a strict reference utility ordering, and restricted solvability, produces `PointwiseAlignmentCertificate P i` directly.

- Discharges: `QuadrupleDescendingSeed` from explicit witnesses + alignment; `PointwiseAlignmentCertificate P i` from the structural inputs alone (no separate strict-pair hypothesis required).
- Realized complexity: 150 lines of new Lean. Six new sorry-free theorems pass the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`).
- Note on the alignment bridge: The original plan suggested coordinate monotonicity as the alignment hypothesis. In the actual mechanization, the additive-representation balance equation supplies the alignment unconditionally once the strict reference utility ordering is given; coordinate monotonicity is implicit in `additiveRep_coordPref_iff`'s use of $\leq$ on $\mathbb{R}$. The strict reference ordering is in turn a one-line consequence of any preference-level strict reference exchange via `quadrupleStrictReferenceUtilityOrdering_of_strictPreference`, so future Wakker-level proofs producing essentiality on k get the alignment free.

Stage T5 (Done). Wakker topological axiom bundle. Stages T1–T4 outputs are now packaged into a single named hypothesis `WakkerTopologicalAxiomBundle R` together with per-coordinate alignment data, plus end-to-end consumers chaining through every previously named layer to the three named raw outputs and to calibrated integer refinement plus full continuity.

- Mathlib used: nothing new beyond Stages T1–T4.
- New defs:

- `CoordinateWakkerAlignmentData R i`: per-coordinate Wakker-monograph data (auxiliary coordinate $k \neq i$, base profile, reference exchange (r, s) with $r \neq s$, bidirectional one-step extensibility, descending and ascending bracketing certificates, and a strict reference utility ordering). This is the precise per-coordinate input Stage T4's alignment theorem consumes.
- `WakkerTopologicalAxiomBundle R`: aggregate body containing `ConnectedCoordinates P`, $\forall i$, `Continuous` $(R.V_i)$, $\forall i$, `CoordinateUtilityUnboundedCertificate` $R\,i$, `SingleCoordinateMonotonic` `P`, and $\forall i$, `CoordinateWakkerAlignmentData` $R\,i$. `RestrictedSolvability P` is kept as a separate explicit hypothesis to match the existing consumer interface.

- New theorems (all sorry-free, all passing the standard axiom audit):
  - `pointwiseAlignmentCertificate_of_coordinateWakkerAlignmentData`: discharge of the per-coordinate alignment certificate from the alignment data via Stage T4's `pointwiseAlignmentCertific`
  - `coordinatePerCoordinateAlignmentCertificate_of_wakkerTopologicalAxiomBundle`: per-coordinate alignment from the bundle.
  - `wakkerMonographTopologyHalfInputCertificate_of_wakkerTopologicalAxiomBundle`: topology half from the bundle, via Stage T3's connectedness + structural monotonicity + continuity + unboundedness route.
  - `wakkerMonographLevelInputCertificate_of_wakkerTopologicalAxiomBundle`: monograph-level aggregate from the bundle, via the existing alignment + topology-half assembly.
  - `wakkerRawOutputs_of_wakkerTopologicalAxiomBundle`: end-to-end discharge to the three named raw outputs.
  - `integerRefinement_and_fullContinuity_of_wakkerTopologicalAxiomBundle`: end-to-end discharge to calibrated integer refinement plus full continuity.
  - `wakkerTopologicalAxiomBundle_of_components`: canonical reverse assembly recording the discharge route a future Wakker-level proof must hit.

- Discharges: collapses the entire remaining open frontier into a single named hypothesis whose body lists exactly the structural-topological inputs Wakker (1989) treats as primitive plus the per-coordinate alignment data. The bundle's only non-trivial residual content is the bracketing-pair certificates inside `CoordinateWakkerAlignmentData`, which Stage T6 will derive from the Archimedean axiom.

- Realized complexity: 150 lines of new Lean. Seven new sorry-free theorems pass the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`).

- Honest scope note: as observed in Stage T4, the additive-representation balance equation supplies the alignment unconditionally once a strict reference utility ordering is given, so coordinate monotonicity does not need to appear as a separate primitive in the alignment data — only in the topology-half side of the bundle (where it discharges full-coordinate continuity via the Stage T3 IVT route). The bundle therefore captures the structural axioms minus the Archimedean-to-bracketing reduction, which is reserved for Stage T6.

Stage T6 (Done). Bracketing-pair discharge from utility unboundedness. The last remaining non-axiomatic piece in Stage T5's bundle: the bracketing-pair certificates inside `CoordinateWakkerAlignm`

Honest scope reduction: under any additive representation, the bracketing inequalities update (update base k s) $(a_0, \text{base}, r) \succcurlyeq$ update (update base k s) i w reduce to bracketing the real value $R.V_i\, a_0 + R.V_k\, r -$

$R.V_k\, s$ between two values of $R.V_i$. This is exactly the content of `CoordinateUtilityUnboundedCertificate R i`, which is already directly provided as a primitive in Stage T5's bundle. The Archimedean axiom appears upstream: it is the structural source of unboundedness through the standard-sequence-pair certificate (`coordinateUtilityUnboundedCertificate_of_strictStandardSequence_pair`, already established), which Wakker (1989) IV.2.6 derives from essentiality + restricted solvability + the Archimedean axiom. Stage T6 mechanizes the analytic core of the bracketing discharge directly under additive representation.

- Mathlib used: nothing new beyond Stages T1–T5.
- New theorems (all sorry-free, all passing the standard axiom audit):
  - `quadrupleDescendingSeedBracketingCertificate_of_unbounded`: under additive representation, the descending seed bracketing inequalities reduce to bracketing the real target value $R.V_i\, a_0 + R.V_k\, r - R.V_k\, s$ between two values of $R.V_i$; coordinate utility unboundedness on i supplies both witnesses v and w directly. The proof uses `R.represents`, `sum_eq_pair_add_rest`, the rest-coordinates agreement, and `linarith`.
  - `quadrupleAscendingSeedBracketingCertificate_of_unbounded`: symmetric to the descending case; the ascending bracketing is the descending bracketing with (r, s) swapped, so it reduces to one application of the descending theorem.
  - `coordinateWakkerAlignmentData_of_connected_continuous_unbounded_strictReference`: end-to-end Stage T6 consumer producing per-coordinate alignment data from connectedness + per-coord continuity + unboundedness + a strict reference utility ordering. The bidirectional one-step extensibility comes from Stage T2; the bracketing certificates from Stage T6 above; the strict reference utility ordering is supplied as a witness.
  - `wakkerTopologicalAxiomBundle_of_strictReferenceWitnesses`: the end-to-end discharge of Stage T5's bundle. Given connectedness, per-coord continuity, unboundedness, structural monotonicity, and per-coordinate strict reference exchange witnesses (the precise content of essentiality on the auxiliary coordinate), the bundle assembles automatically.
  - `wakkerRawOutputs_of_strictReferenceWitnesses`: the end-to-end consumer to the three named raw outputs.
- Discharges: closes the bracketing-pair component of Stage T5's bundle. The remaining structural input is the per-coordinate strict reference exchange witness, which Wakker (1989) supplies through essentiality on the auxiliary coordinate.
- Realized complexity: 200 lines of new Lean. Five new sorry-free theorems pass the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`). This is well under the 400-600 line estimate because the analytic core reduces to `CoordinateUtilityUnboundedCertificate` plus `linarith` on additive sums, rather than the full standard-sequence overshoot argument.
- Honest scope note: the discharge route uses `CoordinateUtilityUnboundedCertificate R i` directly, which Stage T5's bundle takes as primitive. Wakker (1989) IV.2.6 derives unboundedness from the Archimedean axiom + essentiality + restricted solvability via standard-sequence overshoot; that derivation is theorem-backed in the existing artifact via `coordinateUtilityUnboundedCertificate_` which composes with the Archimedean structural axioms cleanly. The Stage T6 theorems above mechanize the precise per-coordinate alignment-data discharge content.

8.3 Cumulative impact

After Stages T1–T6, the entire Wakker IV.2.7 construction is mechanized from *only*:

1. `IsWeakOrder P` and `TradeoffConsistency P` (already typeclass-instances; no work)
2. `PreferenceContinuous P` (Wakker IV.2.3)
3. `ConnectedSpace (X i)` for each coordinate (free for $\mathbb{R}$)
4. `Essential P i` for each coordinate (Wakker III.2.1)
5. `RestrictedSolvability P` (Wakker IV.2.4)
6. `Archimedean P i` for each coordinate (Wakker IV.2.6)

These are exactly Wakker's six structural axioms. `SingleCoordinateMonotonicityAxiom` is no longer a primitive input (Stage T3 derives it); the construction-stack ladder collapses entirely (Stages T4–T6 close every gap below the existing seam).

8.4 Construction-side certificates now discharged

The four items that were open in earlier snapshots are no longer future work. The topology-side stages T1–T6 still supply the consumer half, but the Lean development now closes the construction side as well, so the Wakker / Debreu–Koopmans route is theorem-backed end-to-end.

O1. O1 (`hConstruct`) — discharged. The shared-pivot Step-4 lift is now theorem-backed by `pairwiseSliceRepresentationsAtPivot_of_stage3FiniteCutCoverage_and_normalization`. On top of that, `allPairsAdditivityCertificate_of_stage4PivotSliceRepresentationData`, `wakkerStep5StrictMonotonicityCertificate_of_stage4PivotSliceRepresentationData`, and `wakkerStep5CoordinateImageCoverageCertificate_of_stage4PivotSliceRepresentationData` discharge the three non-circular Stage-5 residuals for one global utility family V. The assembler `wakkerStage5GlobalGluingData_of_stage3FiniteCutCoverage` then feeds `wakkerStage5AdditiveAssemb` yielding `Nonempty` (`AdditiveRep` P) and hence `hConstruct`.

O2. O2 (`hglobal`) — discharged. The same Stage-5 chain supplies the reverse global-additive endpoint. The biconditionals `wakkerStage5GlobalGluingData_iff_wakkerStage5AdditiveAssemblyData` and `wakkerStage5GlobalGluingData_iff_additiveRepHypothesisBundle` certify that the explicit Stage-5 surface is logically equivalent to the global additive target, so the paper no longer relies on an external `hglobal` black box.

O3. O3 (`hConc`) — discharged. The hard quasi-to-concave step is packaged by `quasiToConcaveStrengtheningC` midpoint concavity is isolated as `SliceMidpointConcavityCertificate`, the convex-analysis upgrade is isolated as `MidpointAndContinuityToConcavityResidual`, and the wrapper `twoCoordinateConca` closes the end-to-end two-coordinate concavity consumer.

O4. O4 (`hPairConc`) — discharged. Per-coordinate concavity is now routed through the pivot-indexed bundle `PerCoordinatePairConcavityAtPivotCertificate`. The constructor `perCoordinateConcav` closes the induction via the already-existing base-and-pair transfer, and the composite wrapper `perCoordinateConcavityCertificate_of_pairConcavity_and_coordinateImageCoverage_and_continu` — carrying the M4 continuity surface explicitly as a bystander input — together with `debreu_koopmans_hard_` discharges the Debreu–Koopmans hard direction end-to-end.

Every public theorem in the O1–O4 closure chain still audits only [`propext`, `Classical.choice`, `Quot.sound`].

8.5 Closure sequence realized in Lean

The construction-side closure that landed in Lean is:

$$\text{A4} \to \text{A1} \to \text{A3} \to \text{A2} \to \text{B} \to \text{C}$$

A4 supplies the shared pivot-coordinate utility V_{j_0} for the Step-4 to Step-5 lift. A1 packages all-pairs additivity for one global utility family, and A3 and A2 then discharge the strict-monotonicity and coordinate-image-coverage residuals for that same family, closing O1 and O2 together. Priority B closes the two-coordinate quasi-to-concave strengthening, and Priority C transfers pivot-based pair concavity to full per-coordinate concavity, closing O3 and O4. Combined with Stages T1–T6, this is now the realized end-to-end route in the artifact rather than a projected schedule.

8.6 What was completed in prior sessions

For the audit trail, the 33 sorry-free factoring layers completed in prior sessions are summarized below; full details appear in earlier versions of this section and in `WakkerRoadmap.CertificateChecklist.explicitC`

- **Phase 0–1** (S1–S5): necessity layer, certificate checklist, projection lemmas, M1 Wakker construction certificate, regression wrappers.
- **M2 frontier** (S6–S11): `TradeoffTransferCertificate` factoring, `UtilityValueRealizingEquivalence` corrected predicate, `TradeoffBracketingForallCertificate` from `CoordinateUtilityUnboundedCertificat` surjectivity route, IVT bridge from continuity to surjectivity.
- **Construction frontier reduction** (S12–S22): rational-image coverage, integer seed, integer refinement, rational refinement/bisection, connected-continuity one-step bracket, refined mesh-family certificate, M4 affine-lift routes through strict standard sequences, primitive certificate stronger-output layer.
- **Construction-stack collapse** (S23–S33): `WakkerStandardSequenceConstructionInputCertificate` (S23), structural input bundle (S24), bare-axiom input bundle (S25), seed-construction certificate (S26), joint seed-and-extensibility certificate (S27), monograph-level aggregate bundle (S28), construction/topology halves (S29), per-coordinate alignment minimal core (S30), pointwise alignment per coordinate (S31), per-quadruple predicates (S32), explicit-witness sub-predicates with essentiality bridge (S33).
- **First topology layer** (S34–S35): `oneStepExtensible_of_continuity_unbounded` via Mathlib IVT (the first sorry-free reduction of `OneStepExtensible` to genuine analytic content); explicit seed indifferences from restricted solvability + bracketing pairs; `CoordinateUtilityUnboundedCertificate` from coordinate surjectivity directly.

In total: roughly 130 new sorry-free theorems passing the standard axiom audit (`propext`, `Classical.choice`, `Quot.sound`), spread across 35 factoring layers. These historical layers are the base that the later T1–T6 topology closure and O1–O4 construction-side closure build on.

9 Conclusion

The Lean artifact now proves Wakker IV.2.7 / Debreu–Koopmans end-to-end from Wakker's six structural axioms only. The topology-side stages T1–T6 and the construction-side A4/A1/A3/A2/B/C chain are all theorem-backed, the remaining consumer interfaces are closed by explicit Lean theorems rather than by named placeholders, and the paper's roadmap table has collapsed from an open-program status board into an audit trail of discharged certificates. Every public theorem in the artifact audits only [`propext`, `Classical.choice`, `Quot.sound`], so the final surface is both end-to-end and kernel-audit clean.